



Improved ADHD Diagnosis Using EEG Connectivity and Deep Learning through Combining Pearson Correlation Coefficient and Phase-Locking Value

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Abstract

Attention Deficit Hyperactivity Disorder (ADHD) is a widespread neurobehavioral disorder affecting children and adolescents, requiring early detection for effective treatment. EEG connectivity measures can reveal the interdependencies between EEG recordings, highlighting brain network patterns and functional behavior that improve diagnostic accuracy. This study introduces a novel ADHD diagnostic method by combining linear and nonlinear brain connectivity maps with an attention-based convolutional neural network (Att-CNN). Pearson Correlation Coefficient (PCC) and Phase-Locking Value (PLV) are used to create fused connectivity maps (FCMs) from various EEG frequency subbands, which are then inputted into the Att-CNN. The attention module is strategically placed after the latest convolutional layer in the CNN. The performance of different optimizers (Adam and SGD) and learning rates are assessed. The suggested model obtained 98.88%, 98.41%, 98.19%, and 98.30% for accuracy, precision, recall, and F1 Score, respectively, using the SGD optimizer in the FCM of the theta band with a learning rate of 1e-1. With the use of FCM, Att-CNN, and advanced optimizers, the proposed technique has the potential to produce trustworthy instruments for the early diagnosis of ADHD, greatly enhancing both patient outcomes and diagnostic accuracy.

Keywords ADHD Diagnosis · EEG · Brain Connectivity · Attention-Based CNN · Pearson Correlation Coefficient · Phase-Locking Value

Introduction

Attention Deficit Hyperactivity Disorder (ADHD) is a common neurobehavioral condition that often begins in childhood or adolescence and can continue into adulthood (Fonseca et al., 2008). ADHD is defined by three main characteristics: inattention, hyperactivity, and impulsivity. Although around 5% of the population is affected by ADHD, the symptoms generally manifest before the age of 12 (Spencer et al., 2007). It is estimated that ADHD affects approximately 12.1% of boys and 3.9% of girls among children (Caci et al., 2014; Kessler et al., 2014). Specifically, in the United States, from 2020 to 2022, the prevalence of diagnosed ADHD among children aged 5 to 17 years was 11.3%, according to the National Health Interview Survey. The condition was more common in boys, with a prevalence of 14.5%, compared to 8.0% in girls (Reuben & Elgaddal, 2024).

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ADHD individuals frequently encounter a range of adverse consequences, including social isolation and loneliness, difficulties in establishing and maintaining relationships, as well as persistent academic challenges and underachievement (Danckaerts et al., 2010; Kandemir et al., 2014). Therefore, early detection of ADHD could play a crucial role in enhancing the quality of life for affected individuals (Kooij et al., 2010). It also allows for timely intervention, which can significantly mitigate potential complications. When ADHD is diagnosed early, interventions such as behavioral therapy, educational support, and medication can be implemented sooner (Alves et al., 2024). These interventions can be effective in managing the symptoms of ADHD, which include challenges in maintaining attention, excessive hyperactivity, and impulsive behaviors. By addressing these core symptoms, interventions can significantly improve daily functioning and quality of life for individuals with ADHD.

The diagnostic assessment of ADHD in children is typically carried out by clinical psychiatrists and psychologists. ADHD cannot be diagnosed with a single specific tool; instead, clinicians rely on identifying symptoms such as inattention and hyperactivity. The diagnostic process involves evaluating the diagnostic and statistical manual of mental disorders-V criteria, conducting medical exams, gathering information from schools, interviewing family members, utilizing ADHD rating scales such as the Conners rating scale, and performing neuropsychological assessments (Keith Conners et al., 1998a, 1998b; Nishiyama et al., 2020; Pappas, 2006). Currently, there are no widely used objective assessment tools to support the diagnosis of ADHD and reduce subjectivity. Thus, obtaining a second opinion through a computational approach would not only increase clinicians' confidence in their assessment but also save valuable time.

Diagnosing ADHD based on observing children's behaviors for at least six months and utilizing an electroencephalogram (EEG) to assess children's brain function can aid in early detection of the disorder (Barry et al., 2003). Researchers suggest that ADHD primarily results from the dysfunction of a specific brain area responsible for regulating attention and focus (Tsakou & Drigas, 2022). EEG supplies information on the neural activity within an individual's brain (Giannakeas, 2018) and presents numerous benefits for examining the brain's functional and effective connectivity. These benefits include rapid time response, portability, and cost-effectiveness in data recording. These factors make EEG a suitable choice for investigating how different regions of the brain communicate with each other.

Despite the encouraging results in the literature, there remains a significant gap in achieving an accurate yet simple model for ADHD classification. Current methodologies in connectivity mapping often rely on a single approach, leading to redundancy with duplicated feature values. This redundancy can diminish the efficiency and effectiveness

of classifiers utilizing such connectivity feature maps. By introducing variability within a symmetric connectivity mapping and incorporating informative values, classification performance could be enhanced through better information integration (Ekhlas et al., 2021).

Furthermore, many studies employing feature extraction techniques fail to account for the interrelations between brain regions, extracting features independently (García-Ponsoda et al., 2023; Jaipriya & Sriharipriya, 2024; Mouleeshuwarappabu & Kasthuri, 2023). This oversight results in a trade-off, as these methods do not leverage the full potential of connectivity between regions, nor do they comprehensively capture both linear and nonlinear relationships.

The proposed approach addresses these limitations by integrating fused brain connectivity with linear and nonlinear connectivity measurements, complemented by an attention-based CNN (Att-CNN). This methodology capitalizes on the advantages of each technique, thereby providing a holistic and detailed representation of EEG data. Doing so captures the intricate dynamics of ADHD-related brain activity more effectively, ultimately aiming to enhance classification accuracy while maintaining the model's simplicity.

Therefore, this study attempted to introduce a novel classification method that integrates linear and nonlinear brain connectivity measures with an attention-based CNN, aiming to enhance diagnostic accuracy for ADHD while ensuring the model's non-complexity and practical applicability. Thus, this study has three major contributions as follows:

- 1) Develop a robust framework that integrates linear and nonlinear brain connectivity measures to capture complex neural patterns associated with ADHD.
- 2) Introduce an Att-CNN model to effectively process and classify these connectivity measures, thereby improving the discriminative power of the diagnostic model.
- 3) Evaluate the performance of different optimization techniques, namely adaptive moment estimation (Adam) and stochastic gradient descent (SGD), across various learning rates (LRs) and EEG frequency bands to identify the most effective parameters for accurate ADHD diagnosis.

The rest of the paper is organized as follows. The related literature is summarized next and discussed in Sect. "Related Works". Section "Material & Method" explains the dataset used in this study and also the proposed approach, including preprocessing steps and the proposed classification framework. Section "Experiments" will present the experimental results, including a detailed analysis of the performance of the proposed method across different frequency bands and optimization techniques. Section "Findings" discusses the implications of the findings, compares them with existing literature, and highlights the strengths and limitations of the proposed approach. Finally, Sect. "Conclusions" concludes

the paper with a summary of key findings, potential clinical applications, and suggestions for future research directions.

Related Works

EEG studies utilize various features for analysis and classification. In some studies, brain connectivity measures have been commonly employed to assess connectivity between various brain regions. In this regard, Ekhlasi et al. (Ekhlasi et al., 2022) employed phase transfer entropy (PTE) in conjunction with graph theory to investigate disruptions in brain connectivity associated with ADHD. They utilized EEG signals to construct weighted directed graphs. By analyzing local graph measurements within theta frequency band, achieving an accuracy of 91.2% in distinguishing between subjects with and without ADHD. Ekhlasi et al. (Ekhlasi et al., 2021) also explored the use of directed PTE (dPTE) to measure effective connectivity between electrode pairs in children with ADHD and healthy control (HC). Following this, a multilayer artificial neural network combined with a genetic algorithm for feature selection was applied, which differentiated the θ band between the two groups with an accuracy of 89.7%.

Kiiski and Furlong (Furlong et al., 2021; Kiiski et al., 2020) investigated the utility of functional EEG connectivity as a neuromarker for adult ADHD symptoms. They employed the weighted phase lag index to quantify the connectivity patterns observed in EEG data. Talebi et al. (Talebi & Motie Nasrabadi, 2022) explored the EEG-based effective connectivity in children with ADHD compared to HC employing artificial neural networks for estimating nonlinear causal relationships. Additionally, they employed the direct directed transfer function to analyze connection patterns. By integrating both linear and nonlinear connection measurements, 99% accuracy was achieved in distinguishing between the ADHD and HC groups.

Mazaheri et al. (2010) identified a distinctive impairment characterized by a functional disconnection between the frontal and occipital cortices in children with ADHD. This deficit was observed specifically during attention tasks and was quantified using correlation indices. Abedinzadeh Torghabeh et al. (2023a) utilized phase-lag index and inter-site phase clustering, applying graph theory to extract features. They employed six classifiers to distinguish ADHD from healthy controls, validated using ten-fold cross-validation (CV) and t-tests statistical analysis. In another study (Abedinzadeh Torghabeh et al., 2023b) they used granger causality to streamline EEG analysis by eight electrodes. granger causality indexes across various frequency bands are transformed into color-coded images and processed through an 11-layer convolutional neural network

(CNN). They Achieved 98.50% accuracy for theta frequency subband.

On the other hand, some studies used features other than brain connectivity, such as nonlinear methods and frequency-domain features, to capture the complexity and characteristics of EEG signals. In a recent study, (Alim & Imtiaz, 2023) extracted linear features from the first four frequency sub-bands of EEG signals, training a Gaussian SVM using a ten-fold CV technique, which resulted in an accuracy rate of 93.2% in effectively classifying ADHD individuals from HC. (Ahmadlou and Adeli 2011a, 2011b) utilized synchronization likelihood and fuzzy synchronization likelihood to analyze EEG patterns in ADHD individuals, demonstrating the ability to differentiate ADHD from HCs with 97.1% and 87.5% accuracy, respectively.

Taghi-Beyglou et al. (2022) introduced a method based on convolutional neural networks (CNN) for detecting ADHD using raw EEG signals. The CNN was structured to extract significant information from both spatial and temporal domains, attaining an accuracy of 86.33%. By leveraging feature maps generated by the CNN and applying logistic regression as a machine learning classifier, the method achieved accuracies of 91.16% during training epochs and 95.83% on previously unseen epochs. Mohammadi et al. (2016) extracted fractal dimension, approximate entropy and Lyapunov exponent features from visual task-based EEG recordings. They achieved classification accuracies of 92.28% and 93.65% using features selected by minimum redundancy, maximum relevance and double input symmetrical relevance, respectively, with a multilayer perceptron (MLP) classifier.

Attallah et al. (2024) extracted statistical and spectral features from preprocessed EEG segments. Then, they integrated models such as CNN with bidirectional long short-term memory and gated recurrent unit-transformer. With chi-square and principal component analysis (PCA)-enhanced features resulted in accuracies of 94.50% and 95.59%, respectively. Nishant et al. (2024) employed mutual information to select the most relevant features from both ADHD and HC groups. PCA was utilized to streamline the dataset while retaining essential features, reducing computational complexity. These selected features were then used to train ten diverse classifiers. Combining these classifiers' outputs into a stacked classifier, with the MLP acting as a meta-classifier, they achieved an accuracy of 92.00%.

Natarajan and Madanian (2024) utilized signal noise ratio to select the most relevant features of EEG data and MLP for data classification, employing both leave one out CV and K-fold methods, achieving an 80% accuracy. Nouri and Tabanfar (2024) divided EEG data into 3-s segments and extracted frequency features using Fourier transform per channel. These features were structured into a matrix (channel $\times$ frequency) for input into a

nine-layer CNN, comprising convolutional, maxpooling, and fully connected layers, concluding with a classification output. They optimized performance by analyzing each input variable's classification output with layer-wise relevance propagation, identifying effective channels. Their method achieved a validation accuracy of 94.52%.

Tanko et al. (2022) introduced EPSPatNet86, an EEG-based ADHD detection system using directed graphs and an innovative eight-pointed star pattern for feature extraction. Their model achieved high accuracy of 97.19% with tenfold cross-validation and 87.60% with subject-wise validation. Helgadóttir et al. (2015) devised a statistical pattern recognition approach using a SVM to distinguish between ADHD and typically developing children. They utilized resting-state coherence power features from EEG electrodes located across various sites on the patients' heads, achieving a peak classification accuracy of 76% that was independent of age.

Esas and Latifoglu (2023) utilized the EEG signal, computing its subbands through robust local mean decomposition and variational mode decomposition, feeding fed as input data to a CNN, achieving a classification accuracy of 87% only using a single EEG channel (Fp1).

Material & Method

The proposed framework of this study is illustrated in Fig. 1. A Detailed explanations of each step will be provided in the following sections.

Dataset

This study utilized an open-access EEG dataset (Ali Motie Nasrabadi et al. 2023) that included 121 children, consisting of 61 children diagnosed with ADHD by an experienced child psychiatrist according to diagnostic and statistical manual of mental disorders-IV criteria. Meanwhile, the HC group included 60 children whose medical histories were examined to ensure the absence of psychological disorders, epilepsy, substance abuse, or head injuries. All children were right-handed, with an average age of 9.73 ± 1.76 years. EEG signals were recorded by a digital device (SD-C24, Sholeh Danesh Co., Tehran, Iran) in the Psychology and Psychiatry Research Center at Roozbeh Hospital (Tehran, Iran) and using a 19-channel setup based on the 10–20 standard international system at a sampling frequency of 128 Hz and 16 bits EEG resolution. The experiment was conducted in a quiet room while the children were instructed to reduce their movement.

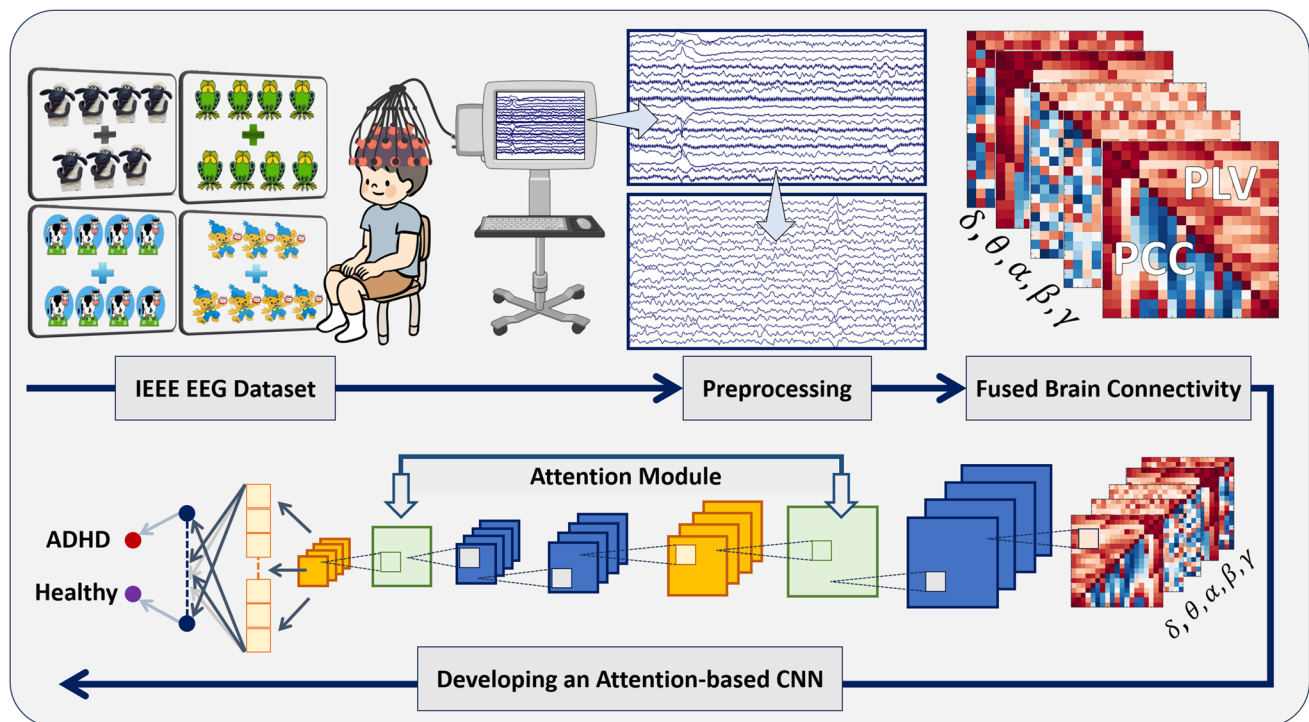


Fig. 1 Proposed framework for classifying ADHD vs. HC involves using fused brain connectivity maps and an attention-based CNN. The study used 19-channel EEG recordings from an existing repository, gathered while participants completed a visual attention task.

Following this, the EEG signals were preprocessed with EEGLAB, and five frequency sub-bands were extracted to build the brain network by assessing functional connectivity using Pearson correlation coefficient and phase-locking value techniques

Given that deficits in visual attention are common in ADHD children, The EEG recording protocol was designed specifically for visual attention tasks. During the tasks, children were shown images featuring engaging characters and instructed to count the number of characters in each picture. Successive images were presented immediately after the child's response to the previous one. The duration of EEG recording was determined by the child's performance, with the correctness of their answers not being taken into account. The EEG recording duration for the ADHD group averages 154.59 s, with a standard deviation of 58.13 s, whereas the HC group has an average duration of 124.91 s and a standard deviation of 29.90 s.

Preprocessing

The raw EEG signals were preprocessed using the EEGLAB toolbox (version 2022.1) (Delorme & Makeig, 2004). First, the EEG signals underwent a high-pass filter at 0.5 Hz to mitigate low-frequency drift. Subsequently, the Clean Raw-data and artifact subspace reconstruction (ASR) plugin (Cleaning Raw Data and Artifact Subspace Reconstruction Plugin 2024; Mullen et al., 2015), utilizing its default settings, automatically removed visible artifacts stemming from electrode displacement. This plugin facilitated the elimination of faulty data channels and suboptimal data segments. It might remove channels exhibiting flat readings or considerable noise, determined by their standard deviation, and those exhibiting poor correlation with other channels (with a correlation threshold of 0.8).

Further in the process, the ASR algorithm (Chang et al., 2018; Chang et al., 2020) was used to identify and eliminate problematic data segments. Subsequently, the deficient/rejected channels were interpolated using the spherical spline method (Perrin et al., 1989), projecting sensor coordinates onto a unit sphere and interpolating signal values for flawed sensor positions based on the readings from functioning locations. Following this, all channels were re-referenced utilizing common average reference values. Then, independent component analysis was applied to decompose the EEG signals, eradicating undesired artifact components like muscular patterns and occurrences of eye blinks or movements frequently observed in visual tasks. The ensuing components were then subjected to recognition and exclusion via the automatic ICLabel plugin (Pion-Tonachini et al., 2019). After that, the EEG signals were filtered into five conventional frequency bands of interest using a finite impulse response filter based on the Kaiser window: Delta (δ :1–4 Hz), Theta (θ :4–8 Hz), Alpha (α :8–13 Hz), Beta (β :13–30 Hz), and Gamma (γ :30–45 Hz)

(Ekhlas et al., 2021; Ekhlas et al., 2022). Subsequently, the EEG signals within each frequency band were segmented into non-overlapping epochs of 768 samples each, equivalent to 6 s.

Linear Nonlinear Connectivity Map

The feature extraction technique transforms input data into new dimensions that represent various combinations (including linear, nonlinear, and directional) derived from the original inputs. The connectivity strength between two electrodes indicates the interaction between two cortical areas during an experiment. This interaction can manifest as direct correlation, inverse correlation, or synchronization, depending on the specific aspects under investigation. Moreover, the nature of these relationships may vary depending on the types of connectivity being analyzed. Two connectivity methods are used for this study: pearson correlation coefficient (PCC) and phase-locking value (PLV). PCC is a linear functional connectivity method, while PLV is a nonlinear one.

PCC

PCC is one of the most frequently utilized measures of linear correlation. This statistical metric gauges the extent to which variables alter their values relative to each other, indicating the degree of linear association between two brain regions (electrodes) of i and j , which can be quantified as:

$$PCC_{ij} = \frac{n \sum E_i E_j - \sum E_i \sum E_j}{\sqrt{n \sum E_i^2 - (\sum E_i)^2} \sqrt{n \sum E_j^2 - (\sum E_j)^2}} \quad (1)$$

where n represents the sample size and E_i and E_j are the individual sample points of channel i and j . Here i and j is 1 to 19. PCC varies between -1 and 1 . A coefficient of -1 signifies a perfect linear inverse relationship between brain regions, 0 denotes no linear correlation, and 1 represents a perfect linear direct relationship between brain regions.

By applying the PCC to EEG data, patterns of functional connectivity between different brain regions can be uncovered. High positive correlation coefficients between brain regions suggest synchronized or coherent activity, indicative of functional connectivity or network synchronization. Negative correlation coefficients in brain regions suggest inhibitory interactions or anti-correlated activity between them (Päeske et al., 2020). Additionally, the magnitude of the correlation coefficient indicates the strength of connectivity observed, helping to characterize neural communication dynamics during cognitive processes like attention, memory, and perception.

PLV

The PLV describes the phase synchronization between two brain regions. It is calculated by averaging the absolute phase differences as follows:

$$PLV_{ij} = \frac{1}{T} \left| \sum_{t=1}^T e^{j(\phi_i^t - \phi_j^t)} \right| \quad (2)$$

where, ϕ^t represents the phase disparity of the signal at time t , i and j denote two brain regions, and T is the total length of the signal. The value of PLV varies between 0 to 1. A PLV value of 0 indicates no discernible connection between the observed signal regions, while a maximum PLV value of 1 signifies a flawless connection. A more detailed description of PLV can be found at (Lachaux et al., 1999).

Fused Connectivity Map

Fusion refers to combining multiple sources of information to create a more comprehensive representation. Fusion techniques are widely used in fields like image processing and medical imaging (Li et al., 2024; SaiKumar & Sulochana, 2024), remote sensing (Ghassemian, 2016), and surveillance systems to improve data quality, accuracy, or robustness. The fusion process, as applied to PCC and PLV brain connectivity maps, involves consolidating essential information from multiple maps into a single, comprehensive map. This fused map surpasses the informativeness and accuracy of any individual source map, capturing all critical details. The goal of this fusion is to reduce the data volume and create more suitable and comprehensible representations for both human interpretation and machine analysis. On the

other hand, enhancing information without increasing size is a significant advantage of fusion over using connectivity maps generated by individual methods.

The Hermes toolbox was utilized for the calculation of connectivity using the PCC and PLV methods (Niso et al., 2013). After generating the PCC and PLV connectivity maps within each frequency band of interest of epoched signals, they are fused into a single matrix. This fusion is feasible because the resulting connectivity maps (PCC and PLV) are undirected and symmetric. Consequently, the fused connectivity map (FCM) has PCC values in the lower triangle and PLV values in the upper triangle. Notably, despite this enhancement, the size of the FCM remains comparable to that of individual methods, making it a cost-effective and efficient strategy for improving information quality in the classification task ahead.

Figures 2, a and b illustrate a connectivity map constructed using PCC and PLV, respectively. It is observed that connectivity mapping often contains redundant information, either partially or completely, such as those produced using methods like PCC and PLV, which display symmetry with mirrored upper and lower portions and partial redundancy with preserved similar values. This redundancy adds computational overhead in classification without corresponding decision-making benefits. In contrast, FCM is advantageous because it integrates upper and lower portions from different connectivity methods, enhancing the information within the feature map. In this figure, the connectivity maps are illustrated in a heatmap wherein the X and Y axes correspond to electrodes, with each pixel's color intensity representing the strength of the connection between each pair. The diagonal pixels signify perfect connectivity for identical electrodes.

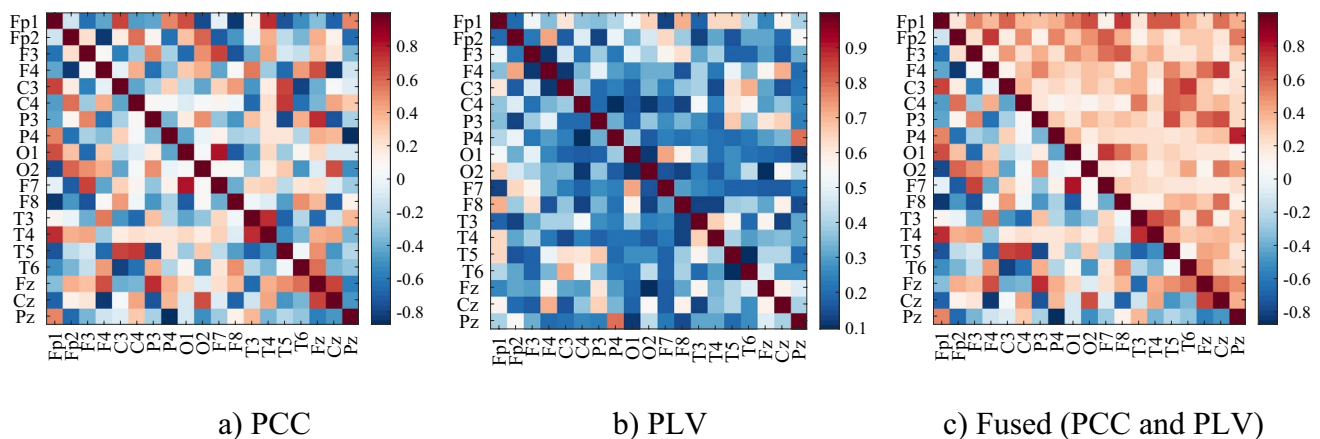


Fig. 2 Sample connectivity maps in one of ADHD subjects in θ sub-band. **a,b** constructed with individual methods, PCC and PLV, respectively. **c** constructed with fused methods (PCC and PLV).

Fused connectivity maps that the upper-right triangle in each map illustrates connectivity computed using the PLV, while the lower-left triangle represents connectivity calculated via PCC

Attention-based CNN (Att-CNN)

Among various deep learning methods, CNNs are the most successful classifiers for two-dimensional data. They can implicitly extract relevant features from input data. A CNN architecture generally comprises an input layer, several convolutional-subsampling layers, a flatten layer, a fully connected layer, and an output layer. The initial operation of a CNN involves convolution performed on the input (matrix, image, or map) using a kernel, resulting in a new convolved matrix. Subsequent subsampling operations downsize the convolved matrix while retaining important features. After one or more convolutional-subsampling operations, the fully connected dense layer processes the output.

This study proposes a novel CNN architecture that leverages multiple convolutional layers, pooling operations, and a channel attention mechanism to enhance feature extraction and improve classification accuracy. In one approach, the channel-attention module is implemented after the first convolutional layer. In another approach, it is embedded after the latest convolutional layer. The second approach is outlined below which is the preferred method in studies that have utilized the attention module. While our work provides a detailed description of the second approach—commonly favored in existing literature—both approaches were developed and evaluated in our study. This ensures that our work explores and tests both methods, highlighting their impact on model performance.

The input layer of the Att-CNN accepts 19×19 FCMs. A 2D convolution operation with a 3×3 kernel, stride size of 1, 32 filters, and the rectified linear unit (ReLU) as an activation function is then implemented. After that, a maxpooling layer with a 2×2 window and stride size of 1 is applied to retain the most relevant features while reducing spatial dimensions. Another 2D convolutional layer follows, with a 3×3 kernel and 64 filters. Then, a deeper convolutional layer with the same kernel size and 128 filters is used.

For the attention mechanism, let's denote the output feature map of the convolution layer as $\mathcal{F}$ and express it as follows:

$$F = X \in \mathbb{R}^{H \times W \times C}.$$

where H , W , and C represent the height, width, and channels, respectively. The channel attention mechanism module is then introduced. This module helps the network to focus on the most informative features by weighting the importance of each channel.

First, the background information from the F is extracted through a layer of average pooling (F_{avg}^c). This layer calculates the average value of each channel across the spatial dimensions (H and W). Given an input tensor F with shape $H \times W \times C$, this operation results in a tensor with shape C ,

where each value represents the average of a corresponding channel. Then, a maxpooling layer calculated the maximum value of each channel across the spatial dimensions, resulting in a tensor with shape C (F_{max}^c). The resultant tensors (F_{avg}^c and F_{max}^c) are then forwarded to a shared network to produce channel attention map ($M_c \in \mathbb{R}^{C \times 1 \times 1}$). The shared network is composed of a MLP with one hidden layer. To reduce parameter overhead, the hidden activation size is set to $\mathbb{R}^{C/r \times 1 \times 1}$, where r is the reduction ratio factor and is considered as 8. After the shared network is applied to each tensors, the output feature vectors are concatenated using element-wise summation. In short, the channel attention is computed as:

$$\begin{aligned} F_{\text{attention}} &= \sigma(\text{MLP}(\text{AvgPool}(F)) + \text{MLP}(\text{MaxPool}(F))) \\ &= \sigma\left(w_1\left(w_0\left(F_{avg}^c\right)\right) + w_1\left(w_0\left(F_{max}^c\right)\right)\right), \end{aligned} \quad (3)$$

where σ denotes the sigmoid function, $w_0 \in \mathbb{R}^{C/r \times C}$, and $w_1 \in \mathbb{R}^{C \times C/r}$. Note that the MLP weights, w_0 and w_1 , are shared for both inputs and the ReLU activation function is followed by $w_0 \cdot F_{\text{attention}}$, which is a tensor that contains the attention weights for each channel of the input F . These weights can scale the original input tensor F to emphasize important channels and suppress less important ones. This process enhances the network's ability to focus on the most relevant features for the task.

Subsequently, the feature maps are downsampled using maxpooling after the channel-attention module, similar to the first pooling layer. The feature maps are then downsampled using max-pooling after the channel-attention module, similar to the first pooling layer (Woo et al., 2018). The pooled feature maps are flattened into a one-dimensional vector. A fully connected layer with 128 neurons and ReLU activation processes the flattened features. This layer captures high-level abstractions. The final dense layer consists of two neurons with softmax activation. It produces class probabilities for the classification of ADHD and HC.

Figure 3 illustrates the proposed attention-based CNN (Att-CNN) model, while Table 1 summarizes its specifications.

Experiments

Experimental Setup

The initial phase of the proposed method, which includes preprocessing EEG signals and calculating FCMs, was carried out using MATLAB software. The following steps involving implementing an Att-CNN were performed on Google Colab using the Keras TensorFlow library and an NVIDIA T4 Tensor Core GPU. The model is validated using

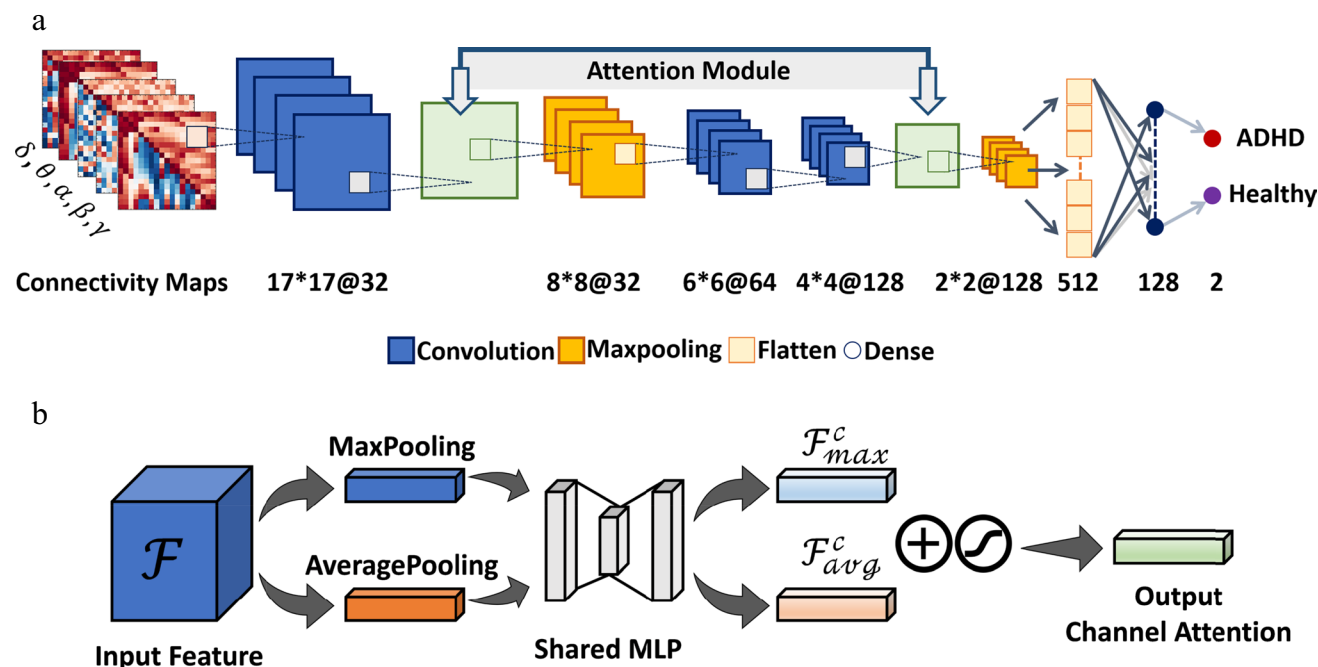


Fig. 3 The illustration of the proposed Att-CNN. To assess the impact of the position of the channel attention mechanism, two different approaches are being considered. In one approach, the channel attention module is implemented after the first convolutional layer. In

another approach, it is embedded after the last convolutional layer: **a** The architecture of the Att-CNN model. **b** The channel-attention module (It adopted from Woo et al. (2018))

Table 1 Specification of the proposed attention-based CNN. The latter approach described below is the preferred method in studies that have used the attention module. In this approach, the attention module is embedded after the final convolutional layer

Layer	Specification	Output Shape
Input layer	-	19 × 19
Convolution	32 Filter, Kernel Size = 3 × 3, Stride = 1	17 × 17 × 32
Maxpooling	2 × 2, Stride = 1	8 × 8 × 32
Convolution	64 Filter, Kernel Size = 3 × 3, Stride = 1	6 × 6 × 64
Convolution	128 Filter, Kernel Size = 3 × 3, Stride = 1	4 × 4 × 128
Channel-Attention	-	4 × 4 × 128
Maxpooling	2 × 2, Stride = 1	2 × 2 × 128
Flatten	-	512
Dense(Hidden)	128 Neurons	128
Dense(Output)	2	2 (ADHD/HC)

the hold-out validation technique. The ADHD and HC subjects themselves were divided into training, validation, and test sets (70%, 10%, and 20%, respectively). After this division, the functional connectivity matrices (FCMs) for each subject were calculated within their respective sets. This approach ensures that no data from any single subject is shared across the training, validation, and test sets, thereby avoiding any data leakage.

Due to the high computational demands associated with cross-validation methods, we opted for the hold-out method in our study. Cross-validation involves extensive model retraining for each fold, which can be both resource-intensive and time-consuming, particularly when optimizing

hyperparameters. The hold-out method, using a 70–10–20 train-validation-test split, was selected for its efficiency in balancing model evaluation with computational feasibility. This approach allowed us to optimize hyperparameters and assess model performance effectively while managing computational resources. Although cross-validation offers a more comprehensive evaluation, the hold-out method provided a practical solution for obtaining reliable results within our resource constraints.

The model's performance is evaluated using accuracy, precision, recall, and F1 Score metrics.

A minibatch size of 128 and a maximum of 250 epochs were used. The early stopping technique is also implemented

to prevent overfitting, which stops the execution if the validation loss does not improve for 50 consecutive epochs. To identify the best optimizer and LR, the model was trained using both Adam and SGD optimizers with LRs ranging from $1e-1$ to $1e-5$ in step of 10. The binary cross-entropy function was used as the loss function. Other hyperparameters not mentioned were set to their default values as specified by the Keras TensorFlow library.

Result & Discussion

This study presented an innovative classification method for children with ADHD and HC using FCM with 3 classifiers: a baseline CNN with no attention module and two attention-based CNNs. For the first part of the analysis, two methods are used for connectivity mapping, PCC and PLV, to capture complex brain-connectivity patterns related to the conventional frequency of interest of ADHD and HC, which the FCMs of ADHD are displayed in Fig. 4.

For the second part of the analysis, to assess the effects of the attention module's placement, it is embedded in two different positions. Firstly, it was embedded after the final convolutional layer and then after the initial convolutional layer. The corresponding results are presented in Table 2. It summarizes the classification accuracy of the baseline CNN and two different approaches to attention placement, Model 1 (with attention applied after the final convolution), and Model 2 (with attention applied after the first convolution). Each model's performance was evaluated across five frequency bands (δ , θ , α , β , γ) using two optimizers, Adam and SGD, at various LRs.

The Baseline model achieved high accuracy with the Adam optimizer, including 97.98% for the δ band, 98.21% for the θ band, and 98.73% for the α band. The model performed well in the θ and α bands across various learning rates. Using the SGD optimizer, the Baseline model showed notable performance, especially in the α band with an accuracy of 98.73%. While some variations in performance were observed with different learning rates, the model consistently demonstrated strong results overall.

Model 1, which includes an attention mechanism after the final convolutional layer, achieved high accuracy with the Adam optimizer, including 98.43% for the δ band and 98.50% for the θ band. The model performed strongly, particularly in the θ and α bands. With the SGD optimizer, Model 1 also showed excellent performance, especially in the θ band with an accuracy of 98.88%. Overall, Model 1 consistently demonstrated strong results across different conditions.

Model 2, which features an attention mechanism after the first convolutional layer, achieved high accuracy with the Adam optimizer, including 98.43% for the δ band and 98.50% for the θ band. The model showed strong

performance, particularly in the θ and α bands. With the SGD optimizer, Model 2 also demonstrated robust results, especially in the θ band with an accuracy of 98.65%. Overall, Model 2 consistently performed well, although some variations in performance were noted under different conditions.

The analysis indicates that both Model 1 and Model 2, which incorporate attention mechanisms, outperform the Baseline model. Specifically, the models perform best with the Adam optimizer at a LR of $1e-3$, achieving the highest accuracy across all frequency bands. Model 1, with attention after the final convolution, slightly outperforms Model 2. The SGD optimizer is less effective at lower LRs for all models. These findings underscore the importance of attention mechanisms and appropriate LRs in enhancing model performance on connectivity maps.

Without attention, the network's performance is highly variable and depends heavily on the optimizer and LR. Adding an attention module after the final convolutional layer stabilizes the performance, allowing the network to better prioritize important features before classification.

Table 3 and Fig. 5 summarize the performance metrics of the models. The Baseline model achieved 98.73% accuracy in the α band with both Adam (stopping at epoch 111) and SGD (converging at epoch 250), showing high precision, recall, and F1 scores. Model 1 reached 98.50% accuracy in the θ band with Adam (training stopped at epoch 95) and 98.88% with SGD (stopping at epoch 122), with improved precision, recall, and F1 scores. Model 2 attained 98.50% accuracy in the θ band with Adam (training stopped at epoch 113) and 98.65% with SGD (finishing at epoch 131), showing strong performance across both configurations.

Figure 6 displays the accuracy curves of the top-performing models referenced in Table 3, illustrating their convergence trends and the epochs at which their training stopped within a maximum of 250 epochs. To do this, an early stopping mechanism was employed during training, causing the models to halt at different epochs to prevent overfitting.

The baseline CNN trained with the Adam optimizer, at a LR of $1e-3$, completed training at epoch 111. This model achieved an accuracy of 98.73%, with precision, recall, and F1 Score of 97.96%, 98.19%, and 98.07% respectively. The accuracy curves for this configuration show a rapid initial increase in both training and validation accuracies, stabilizing around 50 epochs. On the other hand, the Baseline model trained with the SGD optimizer at a LR of $1e-1$ completed training at epoch 250, achieving an identical accuracy of 98.73%, with precision, recall, and F1 Score of 98.40%, 97.73%, and 98.07% respectively. This model exhibited more fluctuations in accuracy during the initial epochs but eventually stabilized, with training and validation accuracies converging around 50 epochs.

Model 1, which incorporates an attention mechanism after the latest convolutional layer, displayed notable

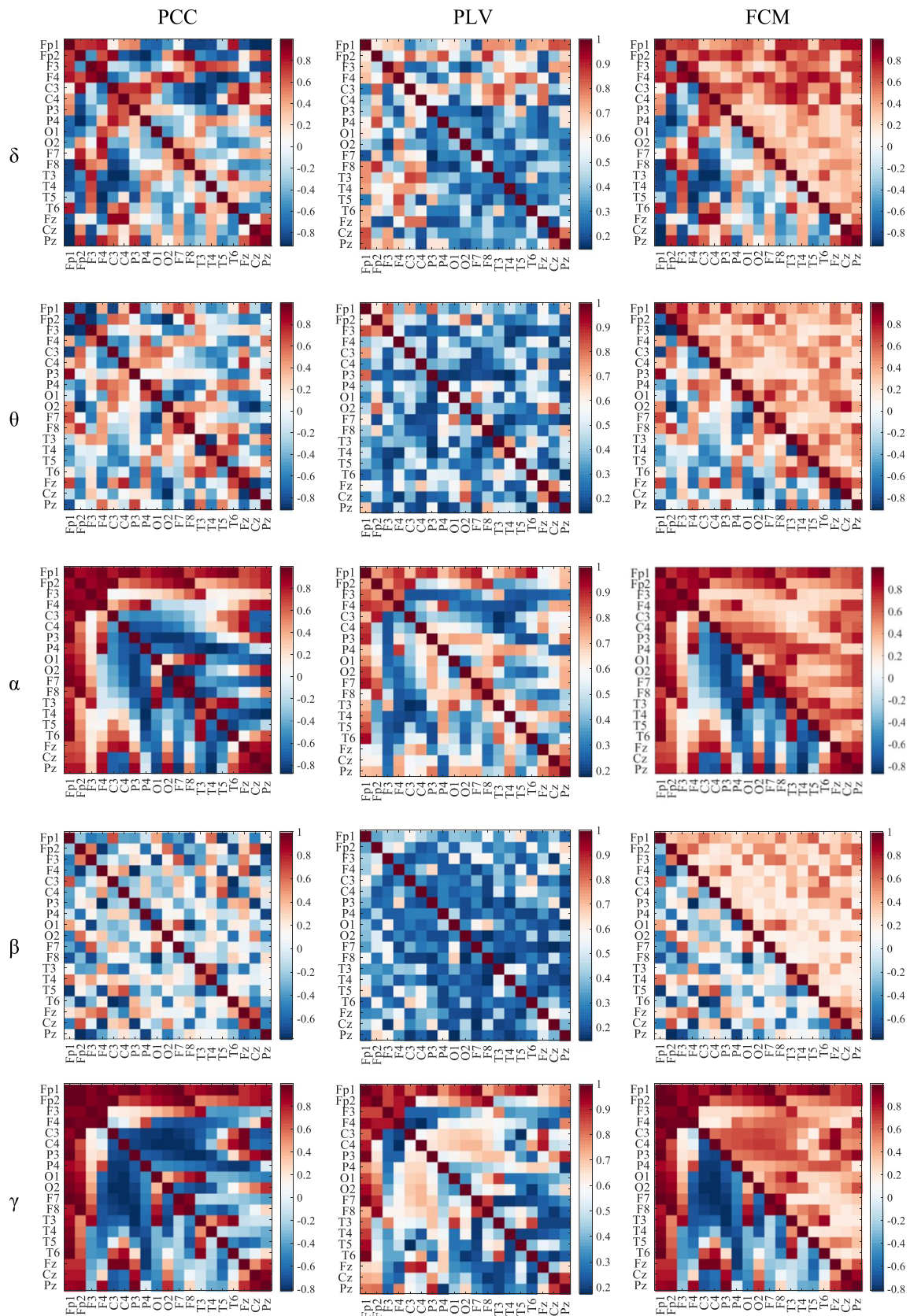


Fig. 4 FCMs that the upper-right triangle in each map illustrates connectivity computed using the PLV, while the lower-left triangle represents connectivity calculated via PCC

Table 2 The classification accuracy (%) of the FCM based on different optimizers across different LRs in five frequency bands of interest

Model	Optimizer	LR	Frequency Bands				
			δ	θ	α	β	γ
Baseline	Adam	1e-1	67.04	67.04	67.04	67.04	67.04
		1e-2	97.31	98.28	95.60	97.01	67.04
		1e-3	97.98	98.21	98.73	98.43	96.04
		1e-4	97.76	97.98	97.68	96.94	92.39
		1e-5	89.33	92.54	87.92	92.54	81.50
	SGD	1e-1	97.91	98.65	98.73	97.98	93.88
		1e-2	94.4	97.39	93.28	97.24	87.54
		1e-3	68.68	82.10	67.04	76.73	67.04
		1e-4	67.04	67.04	67.04	67.04	67.04
		1e-5	67.04	67.04	67.04	67.04	67.04
Model 1	Adam	1e-1	67.04	67.04	67.040	88.36	67.04
		1e-2	96.27	98.43	95.67	96.79	92.91
		1e-3	98.43	98.50	98.21	98.21	96.49
		1e-4	97.76	98.13	97.46	96.27	91.64
		1e-5	85.90	92.46	87.47	91.12	80.46
	SGD	1e-1	97.98	98.88	97.68	98.65	95.97
		1e-2	93.21	97.31	90.67	97.24	87.24
		1e-3	67.04	67.040	67.04	67.04	67.04
		1e-4	67.04	67.040	67.04	67.04	67.04
		1e-5	67.04	67.040	67.04	67.04	33.03
Model 2	Adam	1e-1	67.04	67.04	67.04	67.04	67.04
		1e-2	96.64	95.60	93.51	96.64	67.04
		1e-3	98.43	98.50	98.06	98.13	95.97
		1e-4	97.83	97.16	97.98	98.13	93.73
		1e-5	87.47	91.42	85.60	90.23	80.38
	SGD	1e-1	98.13	98.65	98.06	98.061	93.661
		1e-2	94.03	96.94	91.27	97.16	84.34
		1e-3	67.04	70.02	67.04	67.04	67.04
		1e-4	67.04	67.04	67.04	67.04	67.04
		1e-5	67.04	67.04	67.04	67.04	67.04

Table 3 The evaluation metrics of the best-performing models, indicating their optimizer, LR, band, and the epoch at which training stopped

Model	Optimizer	LR	Epoch Stopped	Band	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
Baseline	Adam	1e-3	111	α	98.73	97.96	98.19	98.07
	SGD	1e-1	250	α	98.73	98.40	97.73	98.07
Model1	Adam	1e-3	95	θ	98.50	98.17	97.28	97.72
	SGD	1e-1	122	θ	98.88	98.41	98.19	98.30
Model2	Adam	1e-3	113	θ	98.50	98.39	97.05	97.72
	SGD	1e-1	131	θ	98.65	98.18	97.73	97.95

performance improvements. When trained with the Adam optimizer at a LR of 1e-3, the model completed training at epoch 95, achieving an accuracy of 98.50%, with precision, recall, and F1 Score of 98.17%, 97.28%, and 97.72% respectively. The accuracy curves indicate a rapid increase in accuracy, stabilizing around 30 epochs, with closely aligned training and validation accuracies, suggesting good

generalization. When using the SGD optimizer at a LR of 1e-1, Model 1 completed training at epoch 122, achieving an accuracy of 98.88%, with precision, recall, and F1 Score of 98.41%, 98.19%, and 98.30% respectively. Despite initial fluctuations, the model's accuracy curves stabilized around 40 epochs, with training accuracy slightly higher than validation accuracy.

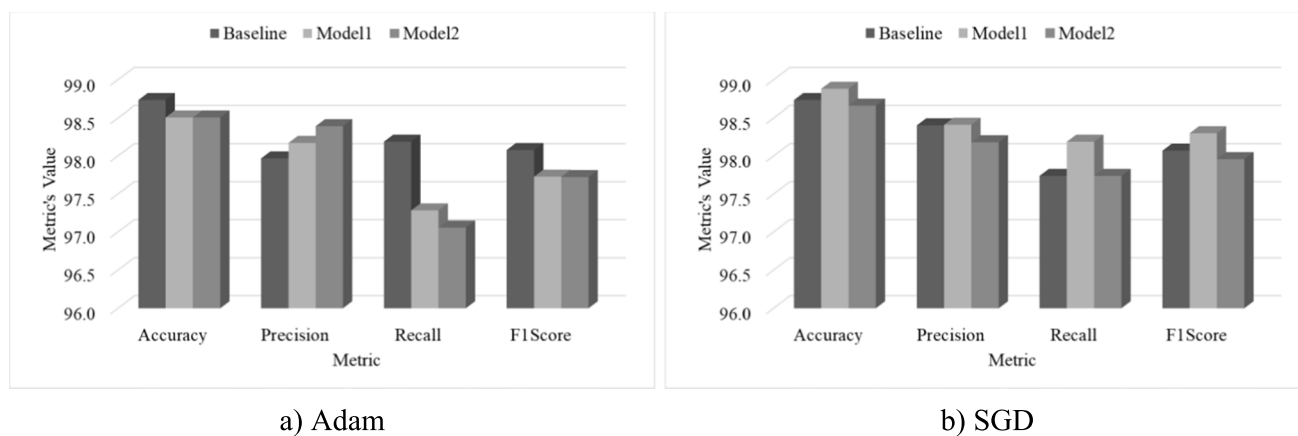


Fig. 5 Comparison of the evaluation metrics of the best-performing models according to the optimizers

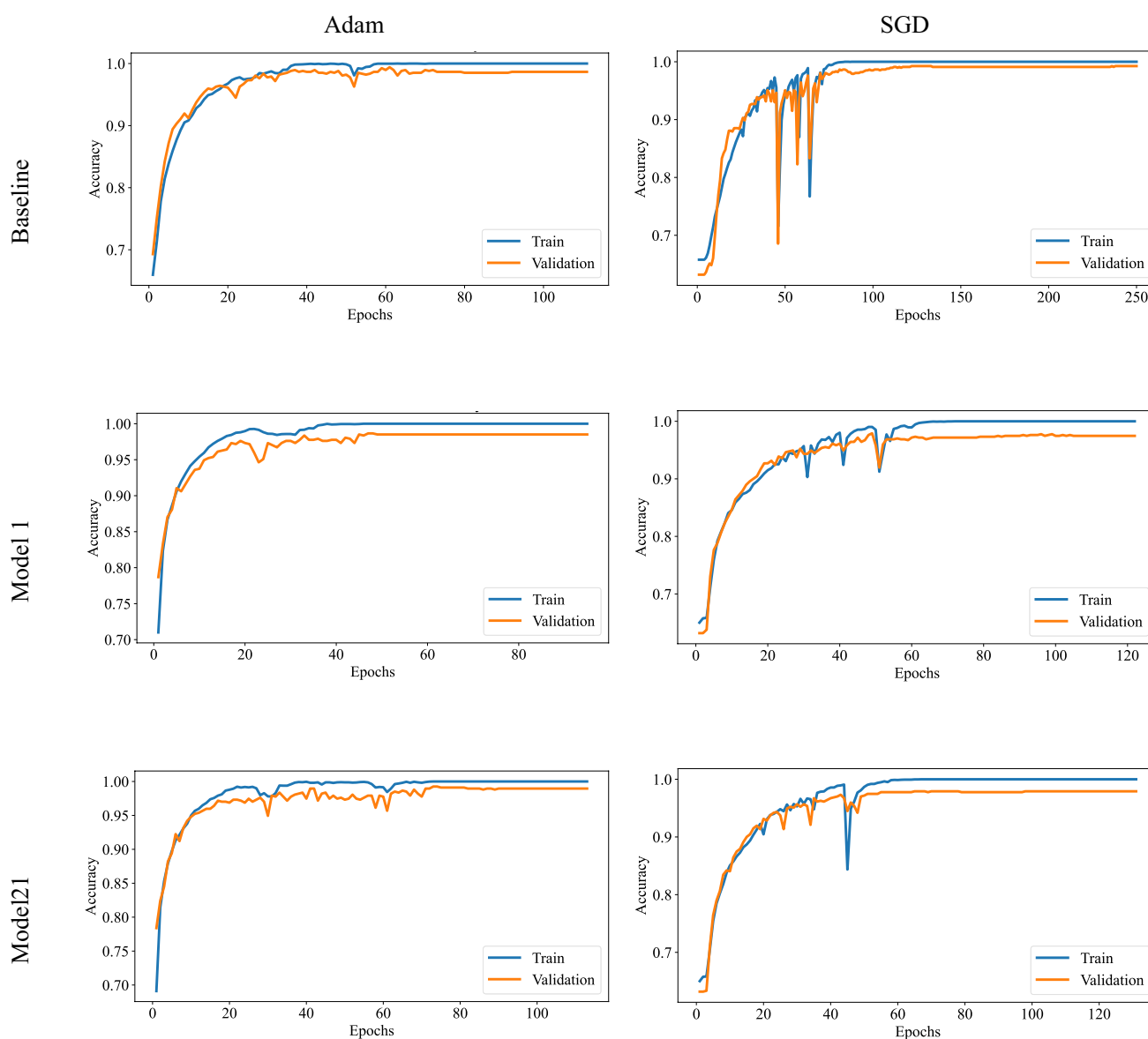


Fig. 6 The accuracy curves of top-performing models which are mentioned in Table 3

Model 2, which integrates attention after the first convolutional layer, also demonstrated enhanced performance. Trained with the Adam optimizer at a LR of $1e-3$, it completed training at epoch 113, achieving an accuracy of 98.50%, with precision, recall, and F1 Score of 98.39%, 97.05%, and 97.72% respectively. The accuracy curves show a smooth improvement, stabilizing around 30 epochs, with closely matched training and validation accuracies, indicating effective training. With the SGD optimizer at a LR of $1e-1$, Model 2 completed training at epoch 131, achieving an accuracy of 98.65%, with precision, recall, and F1 Score of 98.18%, 97.73%, and 97.95%, respectively. Similar to Model 1, the accuracy curves for Model 2 exhibit some initial fluctuations but eventually stabilize around 40 epochs, with closely aligned training and validation accuracies.

The integration of attention mechanism and the choice of optimizers significantly influenced the performance of the CNN models. The Adam optimizer generally resulted in smoother and quicker convergence compared to SGD, which showed more fluctuations but achieved comparable final accuracies. Both Model 1 and Model 2, incorporating attention mechanism, demonstrated slightly better generalization and performance compared to the baseline model. The early stopping technique successfully prevented overfitting across all models and optimizers, as evidenced by the close alignment of training and validation accuracy curves.

Findings

Table 4 compares the proposed method's performance with other state-of-the-art studies that used the same dataset, wherein the proposed model surpassed all of them.

In terms of method, the study (Ekhlesi et al., 2021) utilized dPTE, multilayer ANN, and a genetic algorithm, achieving an accuracy of 89.70% in the θ band. Similarly, (Ekhlesi et al., 2022) employed PTE and graph theory, reporting an accuracy of 91.2% in the θ band. Both methods, while innovative, fall short of the performance demonstrated

by the proposed method, highlighting the efficacy of incorporating attention mechanisms and FCM features.

In 2022, (TaghiBeyglou et al., 2022) focused on spatial and temporal features combined with a CNN, achieving an accuracy of 86.33% across the full frequency band. This study underscores the challenge of effectively integrating complex features into CNN models, a challenge that the proposed method addresses by using attention mechanisms to enhance feature extraction and model performance.

Further, (Nouri & Tabanfar, 2024) utilized CNN with layer-wise relevance propagation in the high γ , achieving an accuracy of 94.52%. Although this method approaches the performance of the proposed method, the difference in the frequency band and the specific methodologies employed highlight the unique advantages of the attention-based approach in the θ band.

The hybrid decomposition-based CNN proposed by Esas and Latifoğlu, (2023) achieved an accuracy of 87% across the full band, demonstrating the potential of hybrid models. However, the simpler structure of the proposed Att-CNN, coupled with focused frequency band analysis, yields better results.

Additional comparisons include the work by Alim and Imtiaz (2023), which utilized linear features and a Gaussian kernel SVM, achieving 93.2% accuracy, and the approach by Nishant and Choi, (2024), which combined mutual information, PCA, and a MLP to achieve 92% accuracy across the full band. Both studies show strong performance, yet the proposed method surpasses them by leveraging advanced feature extraction through attention mechanisms.

Lastly, the method proposed by Natarajan and Madanian (2024) used signal noise ratio for feature selection and an MLP, achieving the lowest accuracy of 80%. This further emphasizes the need for sophisticated feature extraction and selection mechanisms, which are effectively addressed in the proposed method.

The superior performance of the proposed method can be attributed to several factors:

Table 4 Comparison with previous studies that utilized the same dataset

References	Year	Method	Band	Accuracy (%)
Ekhlesi et al., (2021)	2021	dPTE, multilayer ANN, and genetic algorithm	θ	89.70
Ekhlesi et al., (2022)	2022	PTE, and graph theory	θ	91.20
TaghiBeyglou et al., (2022)	2022	spatial and temporal features, and CNN	full band	86.33
Nouri & Tabanfar, (2024)	2022	CNN- layer-wise relevance propagation	high γ	94.52
Esas & Latifoğlu, (2023)	2023	hybrid decomposition-based CNN	full band	87.00
Alim & Imtiaz, (2023)	2023	linear features, and Gaussian kernel SVM	full band	93.2
Nishant & Choi, (2024)	2024	mutual information, PCA and MLP	full band	92.00
Natarajan & Madanian, (2024)	2024	signal noise ratio as a feature selection, and MLP	full band	80.00
Proposed method		FCM and Att-CNN	θ	98.88

Attention Mechanisms: By incorporating attention after specific convolutional layers, the proposed model effectively focuses on the most relevant features, enhancing classification accuracy.

Fused Brain Connectivity Maps: Integrating two connectivity metrics into a FCM provides a richer representation of brain activity, leading to better model performance.

Early Stopping: This technique prevents overfitting, ensuring that the model generalizes well to unseen data.

In terms of physiological interpretability, studies indicated that individuals with ADHD often demonstrate elevated levels of θ activity, particularly in the frontal areas of the brain, in comparison to HC (Clarke et al., 2001; Loo & Makeig, 2012). The θ activity is associated with attention and arousal states, as well as memory and other cognitive processing functions (Chesley et al., 2024). In the context of ADHD, heightened θ activity in the frontal lobes has been associated with challenges in maintaining focused attention, a prominent symptom of the condition. The rise in θ activity could indicate challenges in efficiently filtering distractions and maintaining focus on tasks.

Additionally, some studies have found increased short-range θ connectivity in ADHD but decreased long-range θ connectivity compared to neurotypical controls, suggesting ADHD may involve a disruption in the integration of brain networks in this specific band (Lenartowicz & Loo, 2014). Also, it is reported that the θ band coherence, which measures the synchronization of θ oscillations between brain regions, is atypical in ADHD. Both increased and decreased θ coherence have been reported, depending on the specific brain regions analyzed (Kitsune et al., 2015). Another study's findings suggest that the θ band connectivity can serve as a reliable biomarker for ADHD, with significant differences observed between ADHD and healthy individuals (Abedinzadeh Torghabeh et al., 2023a). Further, it's observed that children with ADHD exhibit increased segregation and decreased integration of brain connectivity in the θ band compared to HCs (Ekhlesi et al., 2022).

Therefore, changes in θ connectivity have been proposed as a potential neurophysiological marker that could aid in the objective diagnosis and monitoring of ADHD. However, more research is still needed to establish the reliability and specificity of theta-based connectivity measures for ADHD detection. This research generally provided one more niche for further exploration in diagnosing ADHD using EEG and machine learning techniques for early detection and diagnosis. Linear and nonlinear connectivity measures are integrated, suggesting the use of complex deep learning approaches to enhance the accuracy and reliability of the developed diagnostic tools. Future work may investigate the applicability of the proposed framework for more extensive and diverse datasets, focusing on the θ band in brain regions.

Additionally, the feasibility of an online, real-time ADHD diagnosis within a clinical setting might be considered.

Conclusions

This study presented a novel framework for classifying ADHD and HCs using EEG data, incorporating fused brain connectivity maps and an Att-CNN. The findings indicate that the attention-based models outperform the baseline CNN, yielding better generalization and overall performance. Specifically, integrating both linear and non-linear connectivity measures allowed us to capture complex brain connectivity patterns more effectively, enhancing diagnostic accuracy.

The application of Adam and SGD optimizers across various LR demonstrated that careful hyperparameter tuning is crucial for optimizing model performance. Additionally, the early stopping technique effectively mitigated the risk of overfitting, as evidenced by the alignment between training and validation accuracy curves.

While the proposed approach demonstrated significant improvement over existing methodologies in the same dataset, indicating that attention mechanisms in deep learning models hold promise for early and accurate ADHD diagnosis, the findings are consistent with previous studies that have identified the theta band as a neuro-marker for ADHD detection. To conclude, the proposed method represents a significant step forward in leveraging advanced EEG analysis and machine learning techniques for ADHD diagnosis, with potential implications for enhancing clinical assessments and interventions.

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Author Contribution EAM: Conceptualization, Writing-Original draft preparation, Reviewing, and Editing. FAT: Conceptualization, Methodology, Software, Writing- Original draft preparation, and Visualization. SAH: Supervision, Investigation, Reviewing, and Editing. MHM: Supervision, Investigation, Writing- Reviewing and Editing, and Validation.

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Data Availability The utilized dataset can be accessed from the IEEE DataPort repository at the following address: [dx.doi.org/https://doi.org/10.21227/rzfh-zn36](https://doi.org/10.21227/rzfh-zn36).

Declarations

Ethics Approval and Consent to Participate Not applicable.

Competing interest The authors declare no competing interests.

Declaration of using Generative AI and AI-assisted Technologies During the preparation of this study, the authors used ChatGPT and Grammarly to improve readability and language. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the publication's content.

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